

TOWARDS ADAPTABLE AND DEPLOYABLE WILDLIFE DETECTION FROM AERIAL IMAGERY

11/28/25

MASc Seminar
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The screenshot displays a software interface for wildlife detection. The main window shows an aerial image of a pond with numerous red and blue dots representing detected wildlife. A purple polygon outlines a specific region of interest. Below the image, there is a legend with labels: 'female duck' (red), 'male duck' (blue), 'rectangle' (grey), and 'Train region' (purple). There are also radio buttons for 'Train only on train region' and 'Train on whole image', and input fields for 'Enter width (float)' and 'Enter height (float)'. At the bottom, there are 'Submit', 'Predict', and 'Train' buttons.

Info History

Annotation Statistics

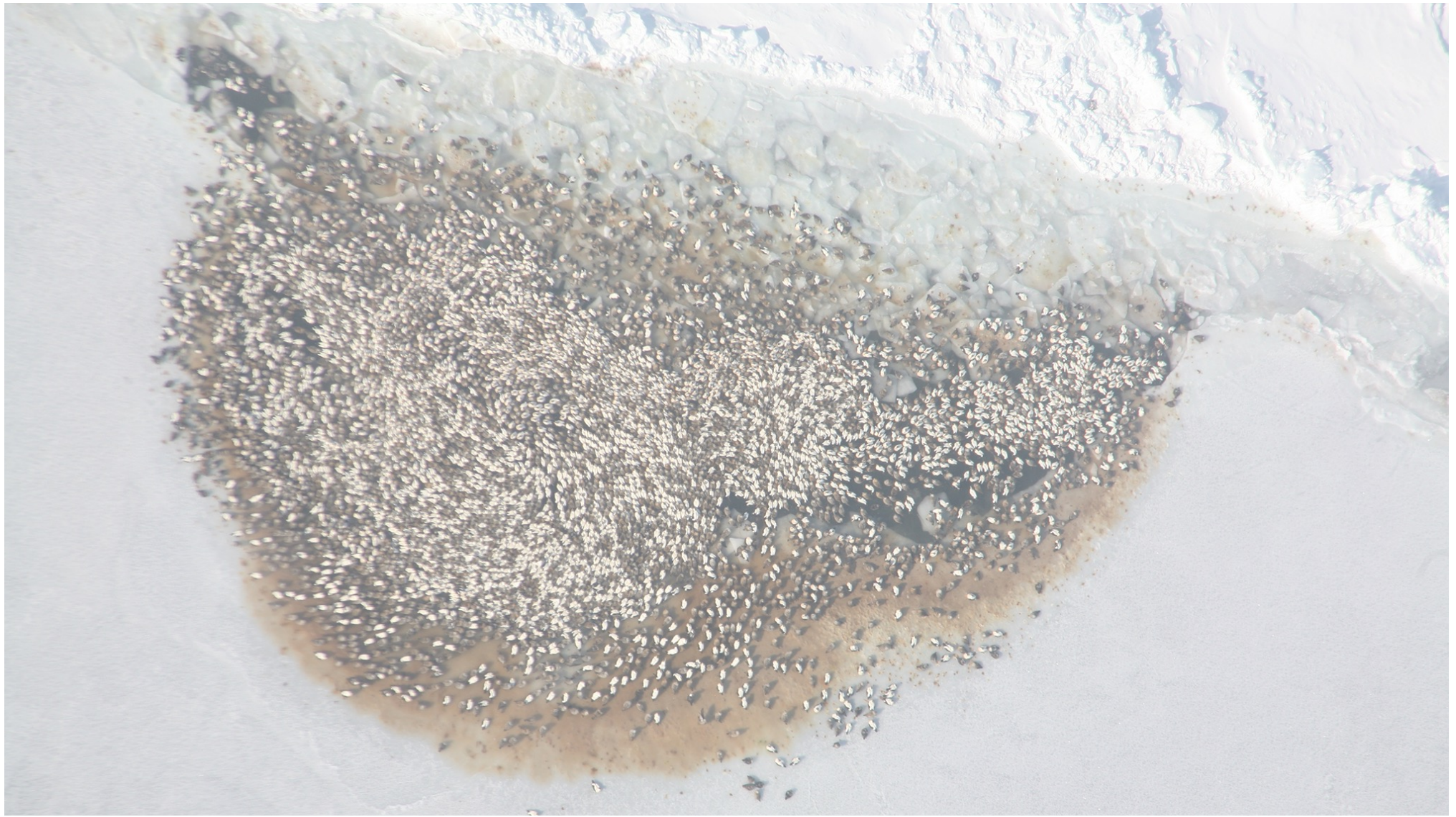
female duck: 1887
male duck: 1881
Train region: 1
Total Valid Annotations: 3769

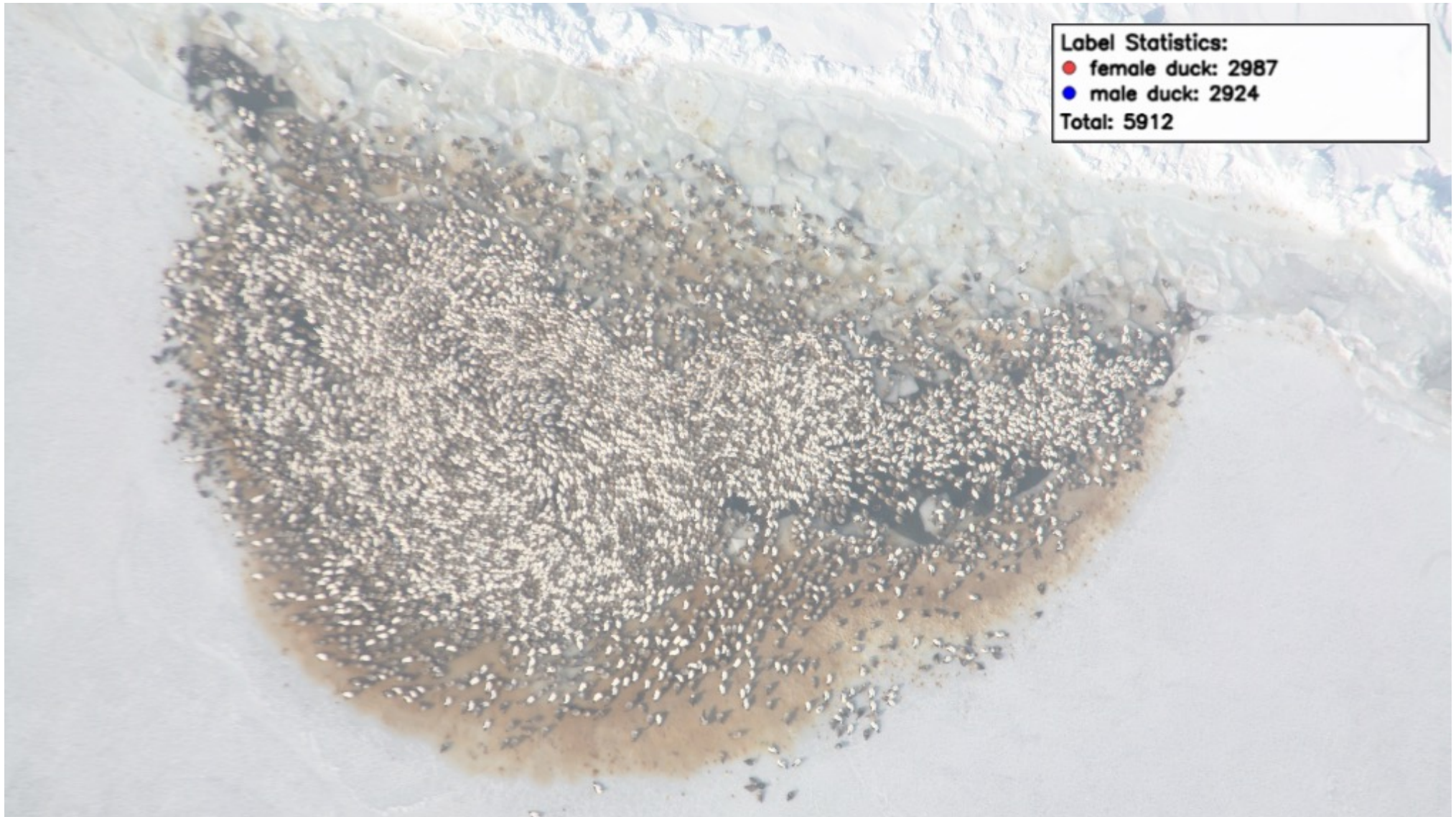
Selection Details

Regions Relations

Manual By Time Conf Thresh

Region	Label	Confidence
1	female duck	0.81
2	female duck	0.79
3	male duck	0.78
4	female duck	0.78
5	male duck	0.77
6	female duck	0.77
7	female duck	0.77
8	male duck	0.77
9	female duck	0.76
10	male duck	0.76
11	male duck	0.75
12	female duck	0.75
13	female duck	0.74





Label Statistics:

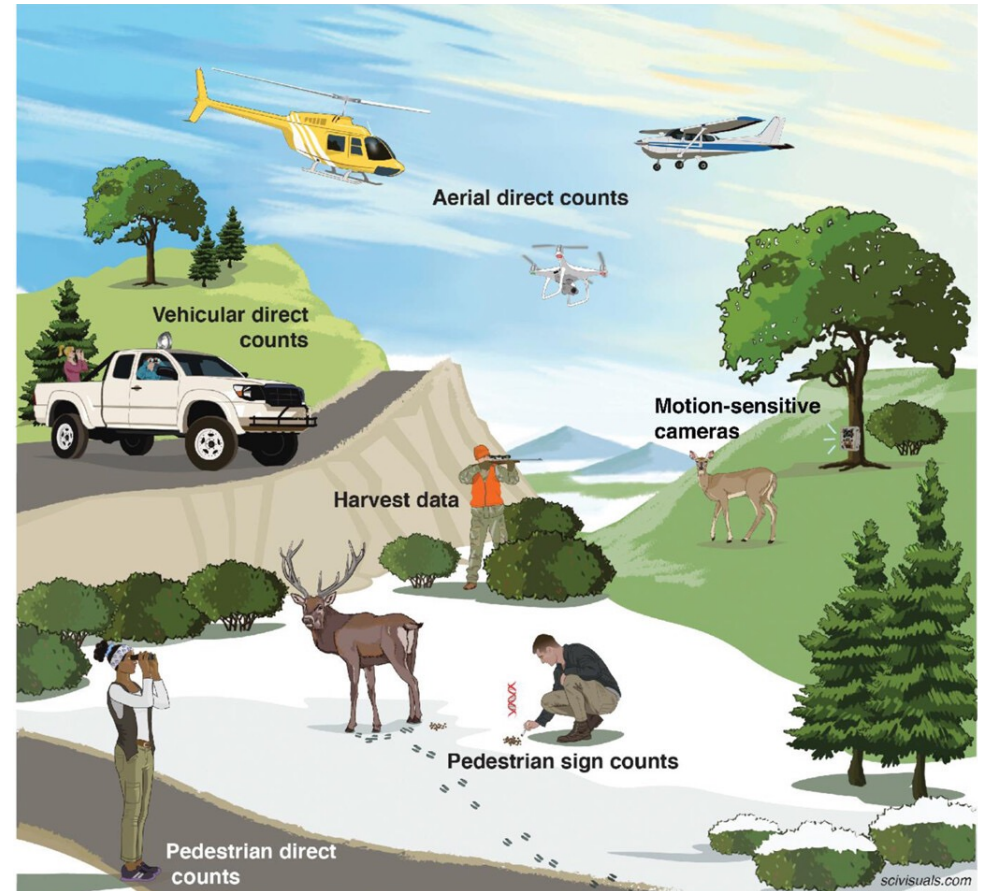
● female duck: 2987

● male duck: 2924

Total: 5912

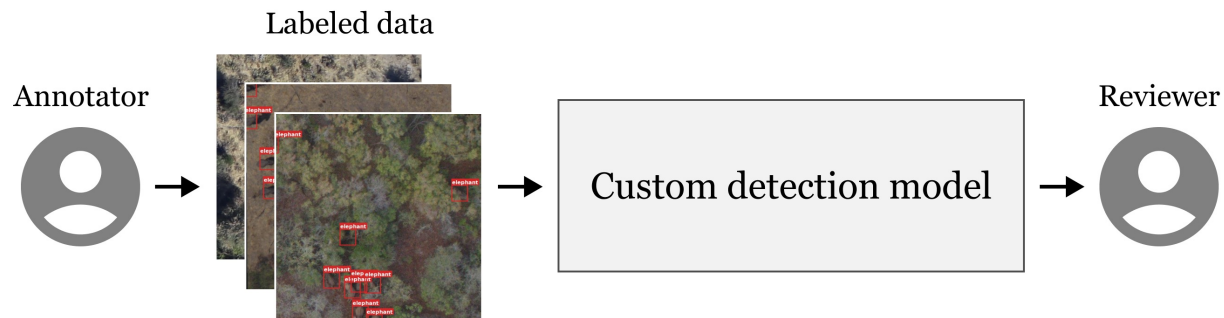
Context

- Accurate wildlife population monitoring is essential for understanding ecological health and guiding conservation efforts.
- Aerial imagery enables large-area monitoring but counting remains labour-intensive and prone to inconsistency.



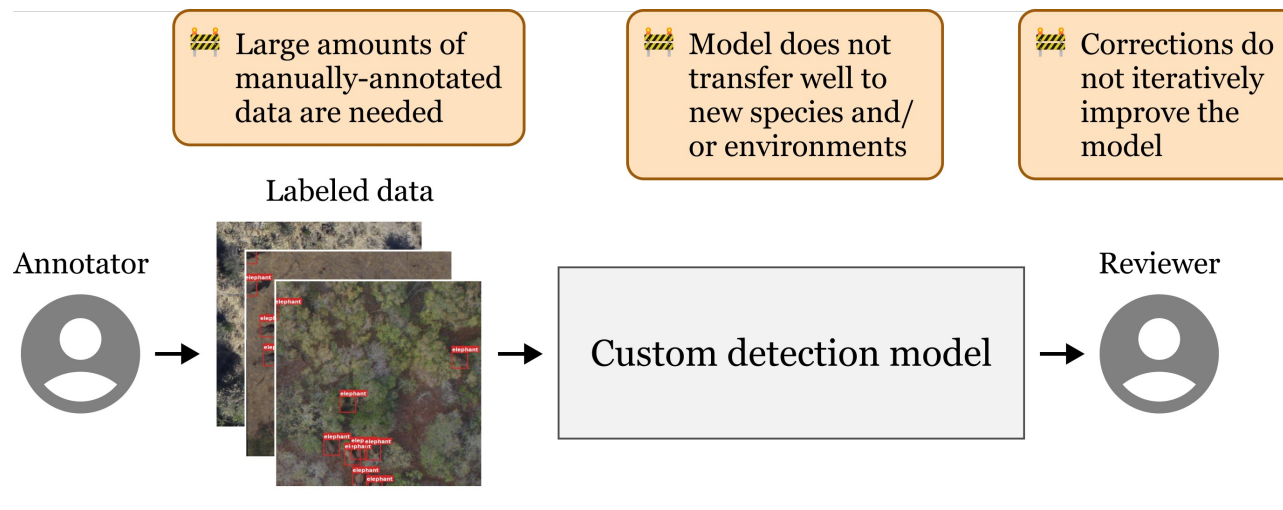
Motivation

- Advancements in computer vision have enabled automated, scalable, and accurate wildlife counting.



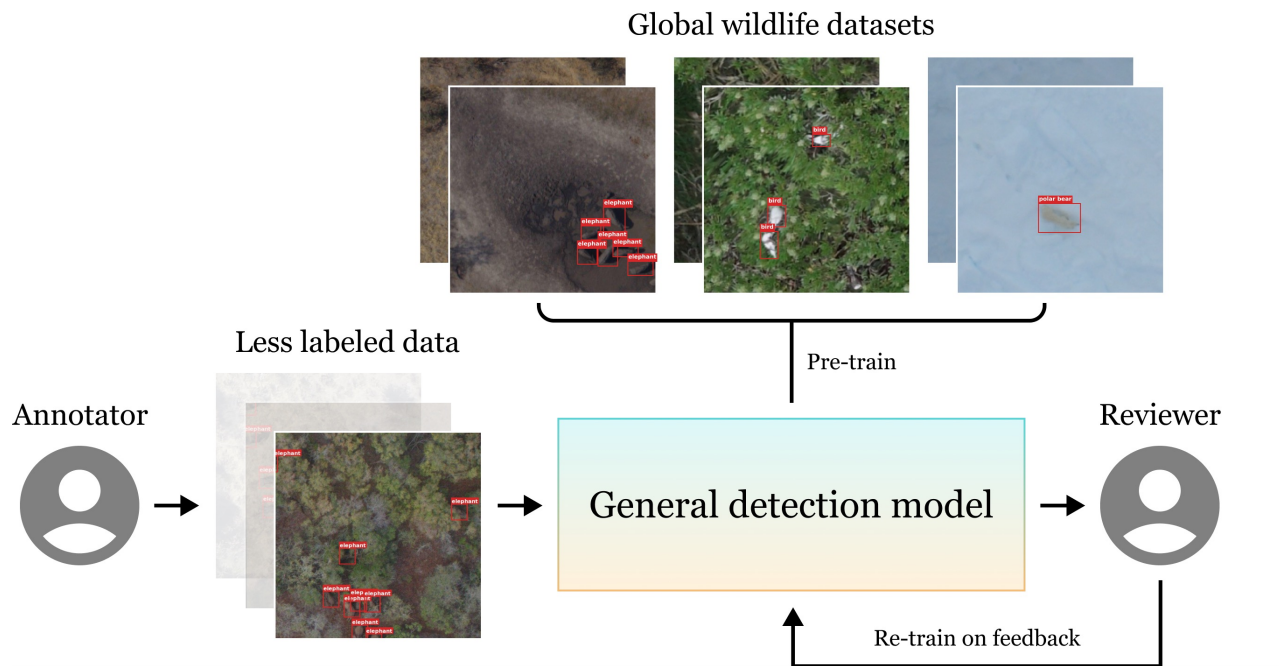
Motivation

- Advancements in computer vision have enabled scalable, accurate, and automated wildlife counting.
- However, several challenges remain:



Opportunities for Improvement

- In this seminar, I'll be proposing a holistic solution to these problems.



Contributions:

1. A multi-species, language-enabled wildlife detection model supporting open-vocabulary detection.
2. A human-in-the-loop annotation workflow allowing incremental fine-tuning from user corrections.
3. A case study with the Arctic Eider Society demonstrating the effectiveness of these components in real-world ecological monitoring.

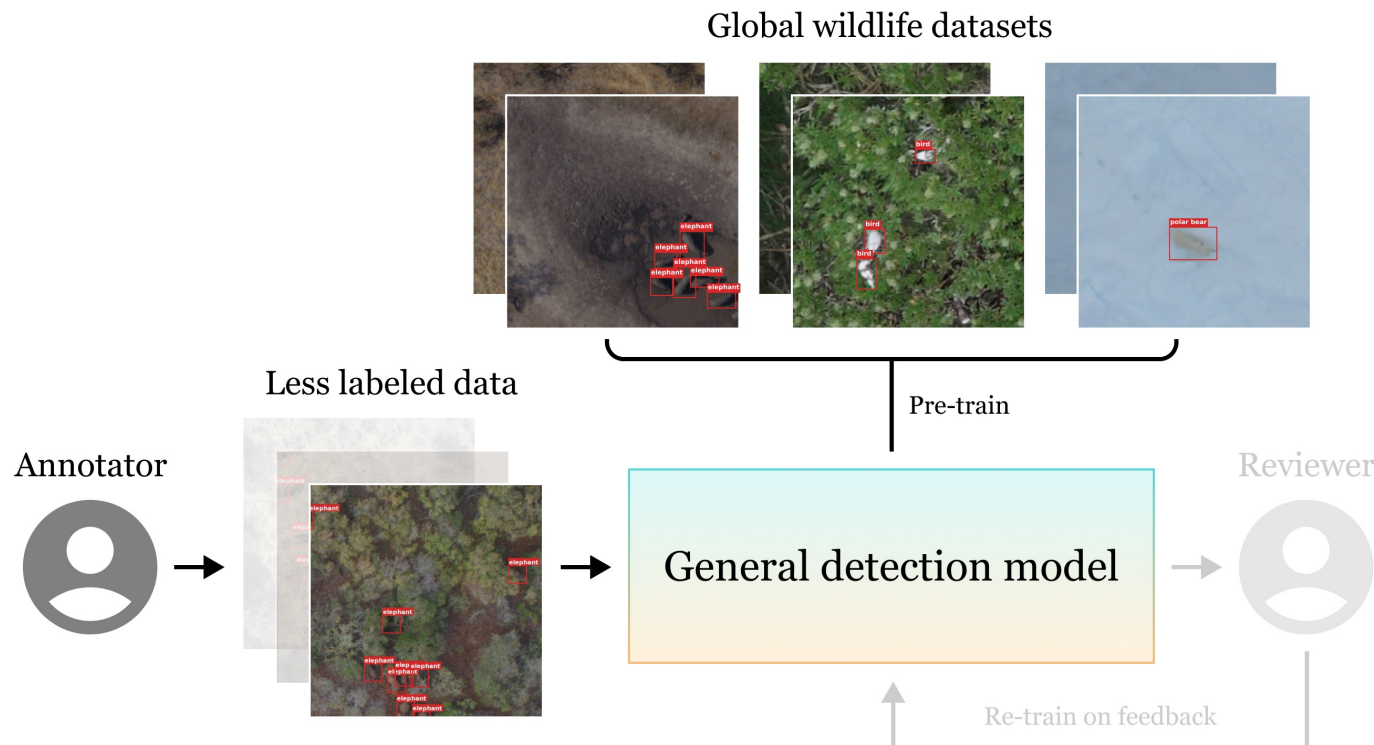
CONTRIBUTION 1

A multi-species, language-enabled wildlife detection model
supporting open-vocabulary identification.

Collaborators: Muhammed Patel, Javier Noa Turnes

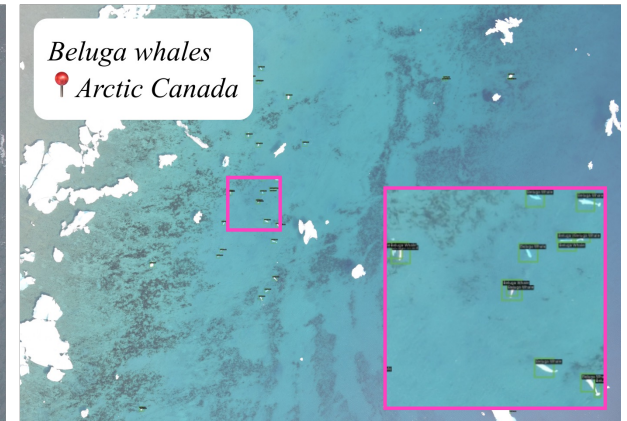
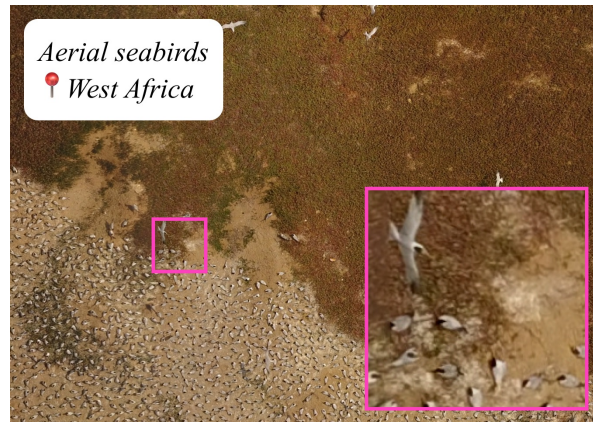
Research Question

- How might we leverage cross-modality embeddings and pretraining on global wildlife datasets to create a single detection model that generalizes well to novel or few-shot species and environments?



Dataset

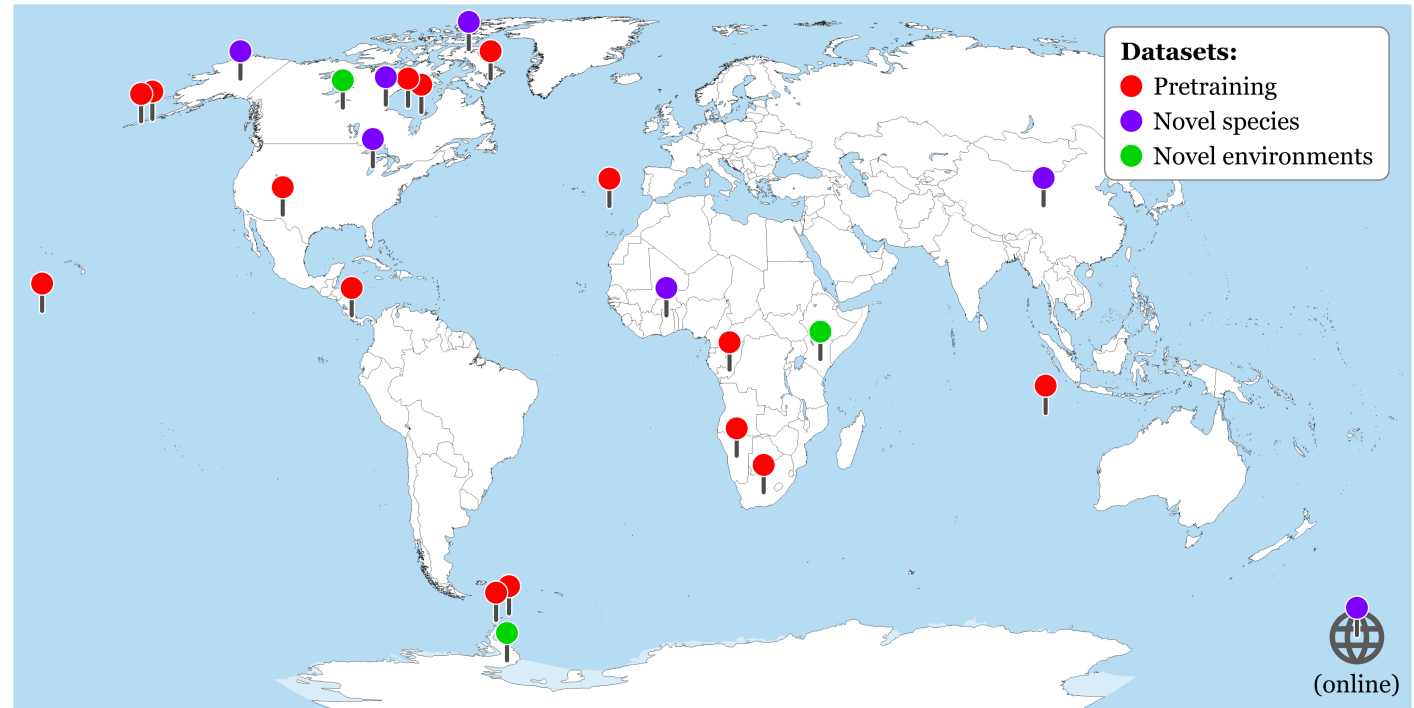
- Consolidation of 25 publicly-available aerial wildlife datasets.



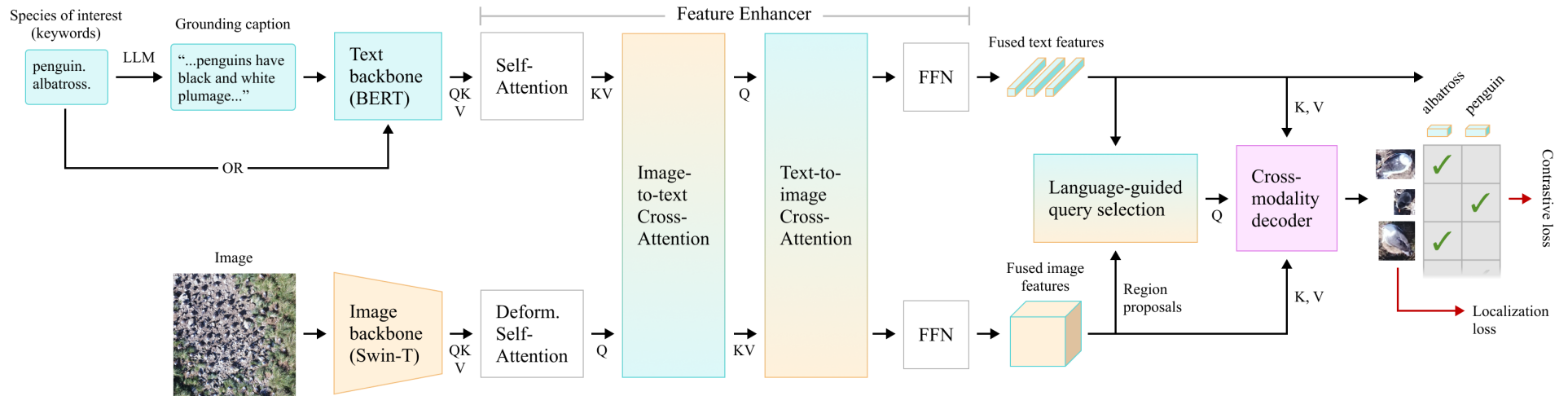
Dataset

- Partitioned into:
 - 15 for pretraining
 - 7 for evaluation on novel species
 - 3 for evaluation on novel environments

	Species labels	# of annotated images	# of annotations
Train	Alcelaphinae, beluga, bird, brant, buffalo, Canada goose, elephant, emperor goose, kob, penguin, sea lion, turtle, warthog, waterbuck	27,684	977,428
Test (novel species)	Camelus, cattle, gray headed gull, great cormorant, great white pelican, gull, kiang, narwhal, polar bear, royal tern, seal, sheep, slender-billed gull, yak, zebra	22,028	320,381
Test (novel environments)	Beluga whale, bowhead whale, elephant, giraffe, narwhal, penguin, zebra	1,226	8,360



Modeling



$$\mathcal{L}_{total} = \lambda_{cls} \mathcal{L}_{cls} + \lambda_{L1} \mathcal{L}_{L1} + \lambda_{GIoU} \mathcal{L}_{GIoU}$$

Keyword	Sentence
Beluga whale	Species that may be present: beluga whales. Belugas are characterized by their predominantly white coloration, which can appear almost ivory in bright lighting. They have rounded heads and a lack of dorsal fin, giving them a distinctive streamlined shape. In water, they may exhibit a subtle bluish or grayish hue, especially in deeper areas. Their bodies feature smooth, gently curving contours, and they may display light markings that are more prominent on their backs.

Experiments

- Pre-trained on eight A100-40GB GPUs for 20 epochs $\approx$ 20 hours
- Training and inference on 1024×1024 patches, merged predictions to produce full-scale images
- Metrics included:

RMSE ↓
(how well did it count?)

$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$

F1 ↑
(how well did it classify?)

Precision

$$\frac{tp}{tp + fp}$$

Recall (Sensitivity)

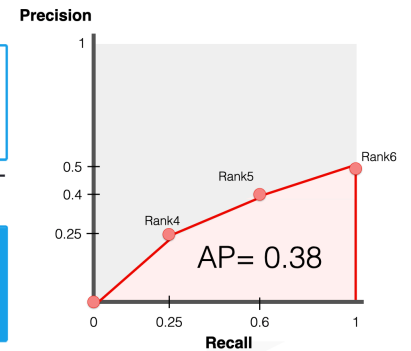
$$\frac{tp}{tp + fn}$$

F1-score

$$\frac{2 * precision * recall}{precision + recall}$$

mAP50 ↑
(how well did it localize and classify?)

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$



Results – Literature Comparison (Pre-training Datasets)

🤔 Is our single OpenWildlife model competitive with several existing models in literature when trained on the same data splits?

✅ We outperform most existing methods, except for the Qian and Virunga-Garamba datasets.

Dataset	Literature Results	OW Keyword	
		Lit. Metric	mAP50
IndOcean [60]	0.72 F1 [60]	0.80 F1	0.833
NewMex [60]	0.76 F1 [60]	0.826 F1	0.728
Palmyra [60]	0.72 F1 [60]	0.78 F1	0.789
Pfeifer [46]	0.66 F1 [60]	0.86 F1	0.842
Seabird	0.72 F1 [60]	0.93 F1	0.959
Qian [47]	66.27 RMSE ‡	85.32 RMSE	0.671
WAID [40]	0.98 mAP50 [40]	0.98 mAP50	0.981
AED [41]	0.89 mAP † [41]	0.90 mAP †	0.856
Virunga-Gar.[7]	0.85 mAP50 [25], 0.835 F1 [8]	0.72 mAP50, 0.85 F1	0.719
Sea Lion [21]	10.86 RMSE [21]	8.13 RMSE	0.662
Turtle [16]	0.27 F1 ‡	0.38 F1	0.26
Izembek [61]	-	-	0.343
DFOW14	-	-	0.856
DFOW15	-	-	0.82
DFOW17	-	-	0.857

Results – OW Keyword vs OW Sentence

🤔 Do LLM-generated captions help OpenWildlife’s ability to detect species?

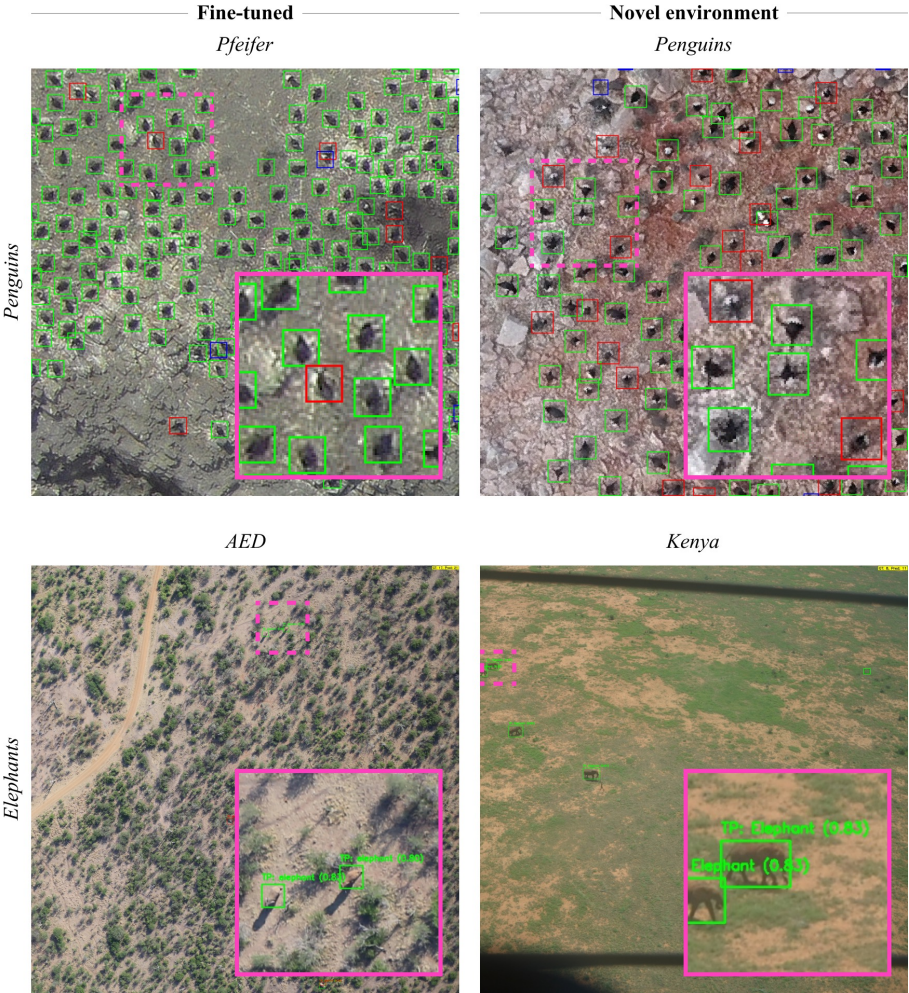
🙋 In certain situations, captions can improve classification accuracy. However, this is inconsistent.

Dataset Name	OW Keyword	OW Sentence
IndOcean	0.833	0.852
Izembek	0.343	0.401
NewMex	0.728	0.642
Palmyra	0.789	0.801
Seabird	0.959	0.957
Pfeifer	0.842	0.83
Qian	0.671	0.639
Sea lion	0.662	0.668
Turtle [16]	0.26	0.340
DFOW14 [36]	0.856	0.807
DFOW15 [39]	0.820	0.800
DFOW17	0.857	0.837
AED [41]	0.858	0.864
SAVMAP	0.379	0.401
Viruga-Gar	0.719	0.719

Results – Novel Environments

- 🤔 How does our OpenWildlife model perform on *environments* that it hasn't seen in training?
- ✅ Up to 0.733 mAP50, surpasses literature results when fine-tuned.

Dataset	Novel Environments		Lit. Results	Finetune	
	Sentence (mAP50)	Keyword (mAP50)		Lit. Metric	Keyword (mAP50)
Penguin	0.733	0.619	-	-	0.755
DFOW23	0.024	0.362	-	-	0.435
Kenya	0.384	0.326	0.77 mAP30 [12]	0.869 mAP30	0.854



Results – Novel Species

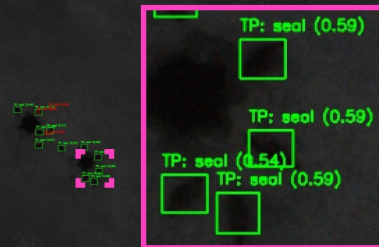
🤔 How does our OpenWildlife model perform on *species* that it hasn't seen in training?

✅ Up to 0.597 mAP50, competitive with literature results when fine-tuned. Model exhibits semantic understanding.

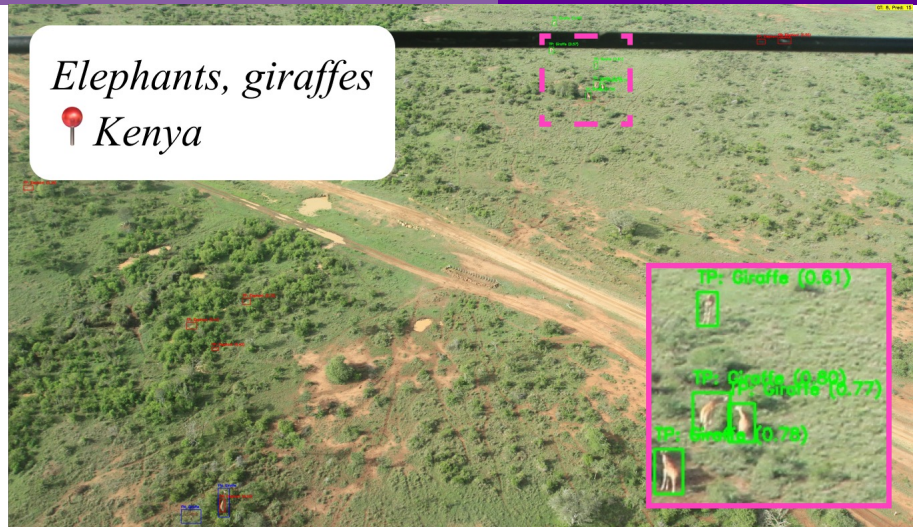


Dataset	Novel Species		Lit. Results	Finetune	
	Sentence (mAP50)	Keyword (mAP50)		Lit. Metric	Keyword (mAP50)
WAfrica	0.465	0.533	0.79 F1 [23]	0.91 F1	0.908
Michigan	0.179	0.131	0.72 F1 [60]	0.82 F1	0.813
DFOW16	0.576	0.597	-	-	0.805
Polar Bear	0.369	0.224	-	-	0.716
ArctSeal	0.257	0.183	0.87 F1 ‡	0.87 F1	0.916
Han	0.106	0.161	0.89 mAP50 [18]	0.96 mAP50	0.964
WAID	0.229	0.212	0.98 mAP50 [40]	0.97 mAP50	0.967

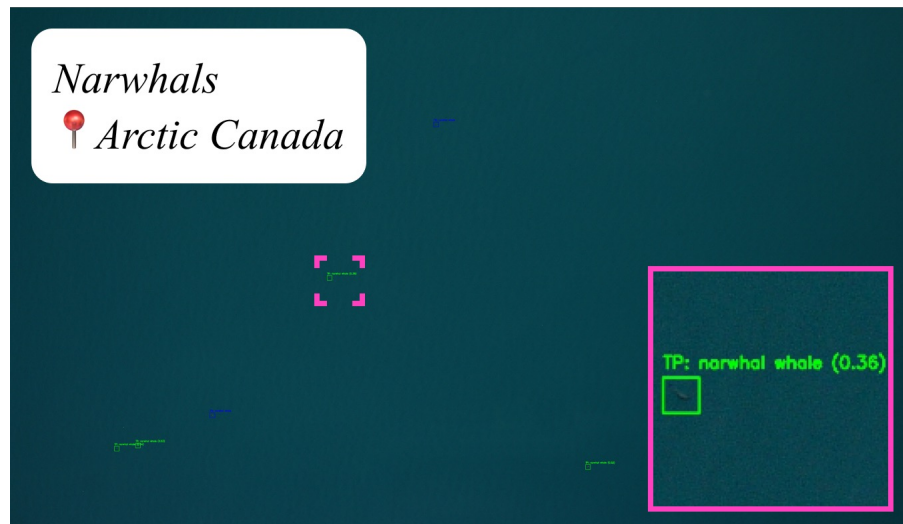
Arctic seals
📍 *Alaska*



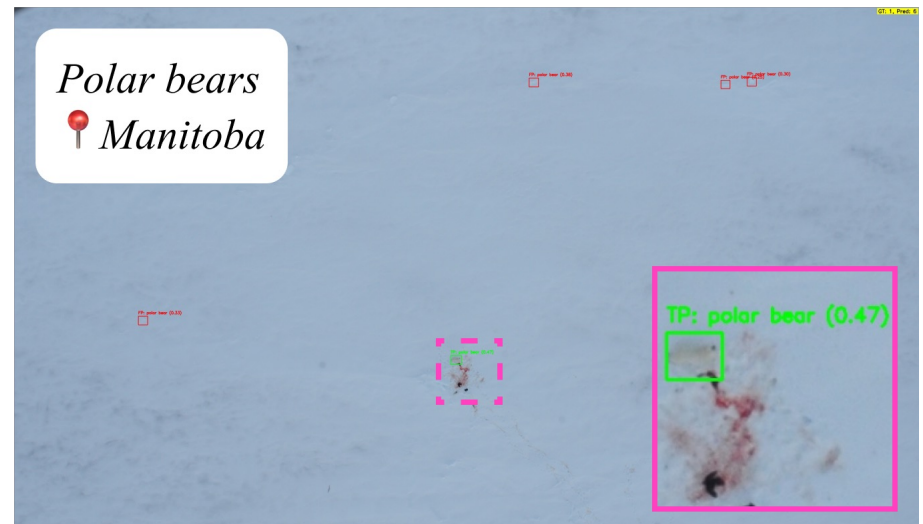
Elephants, giraffes
📍 *Kenya*



Narwhals
📍 *Arctic Canada*



Polar bears
📍 *Manitoba*



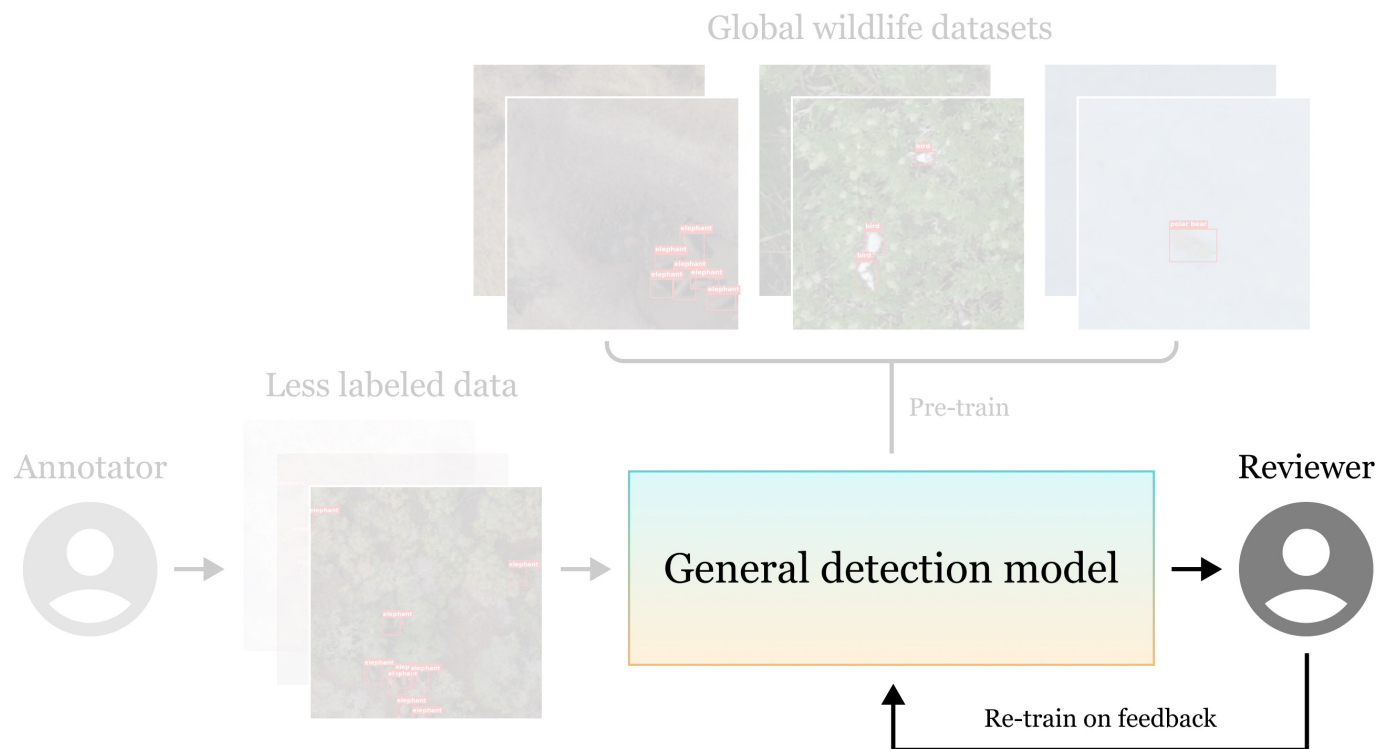
CONTRIBUTION 2

An annotation workflow allowing incremental fine-tuning from user corrections.

Collaborators: Aryan Kalia, Zhonghao Zhang, Hudson Sun,
Muhammed Patel

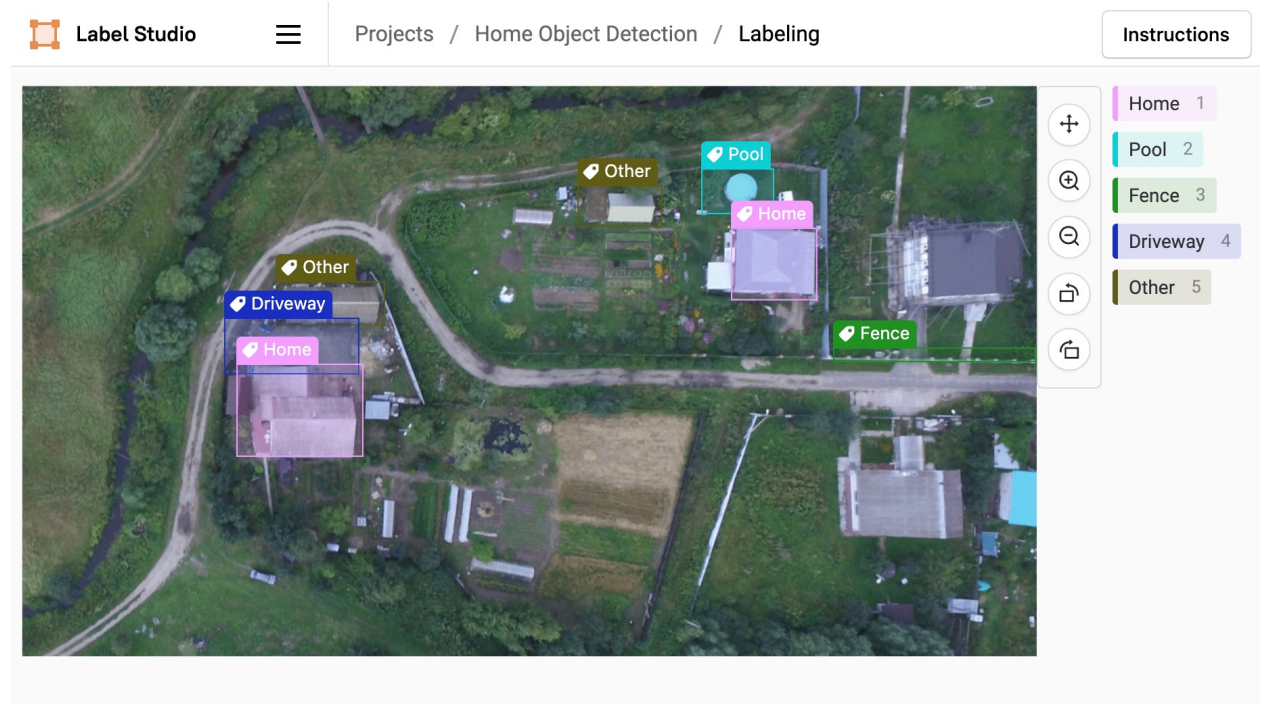
Research Question

- How might we design a system where a model learns incrementally from human correction on aerial imagery of high-density wildlife?

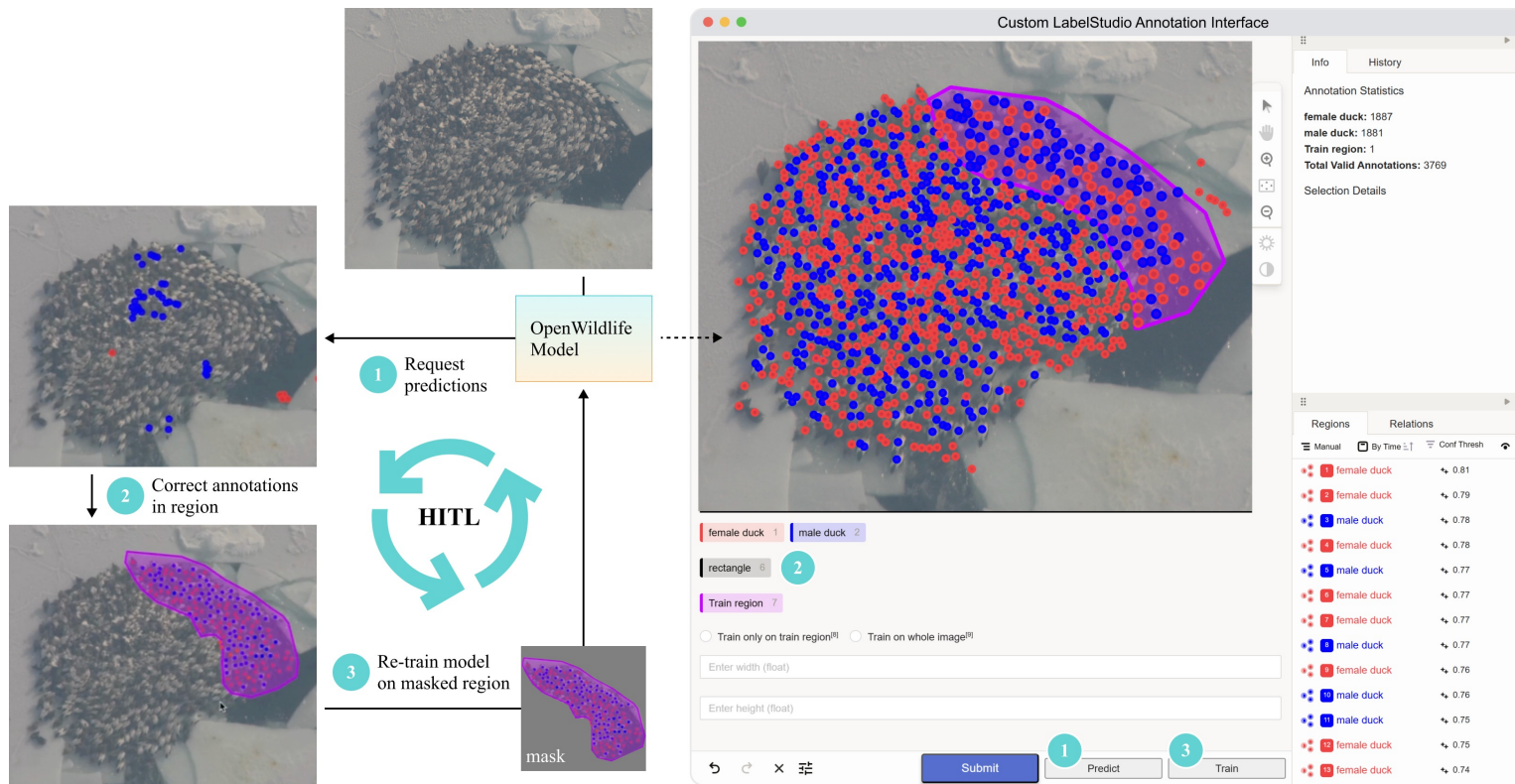


Limitations with Existing Systems

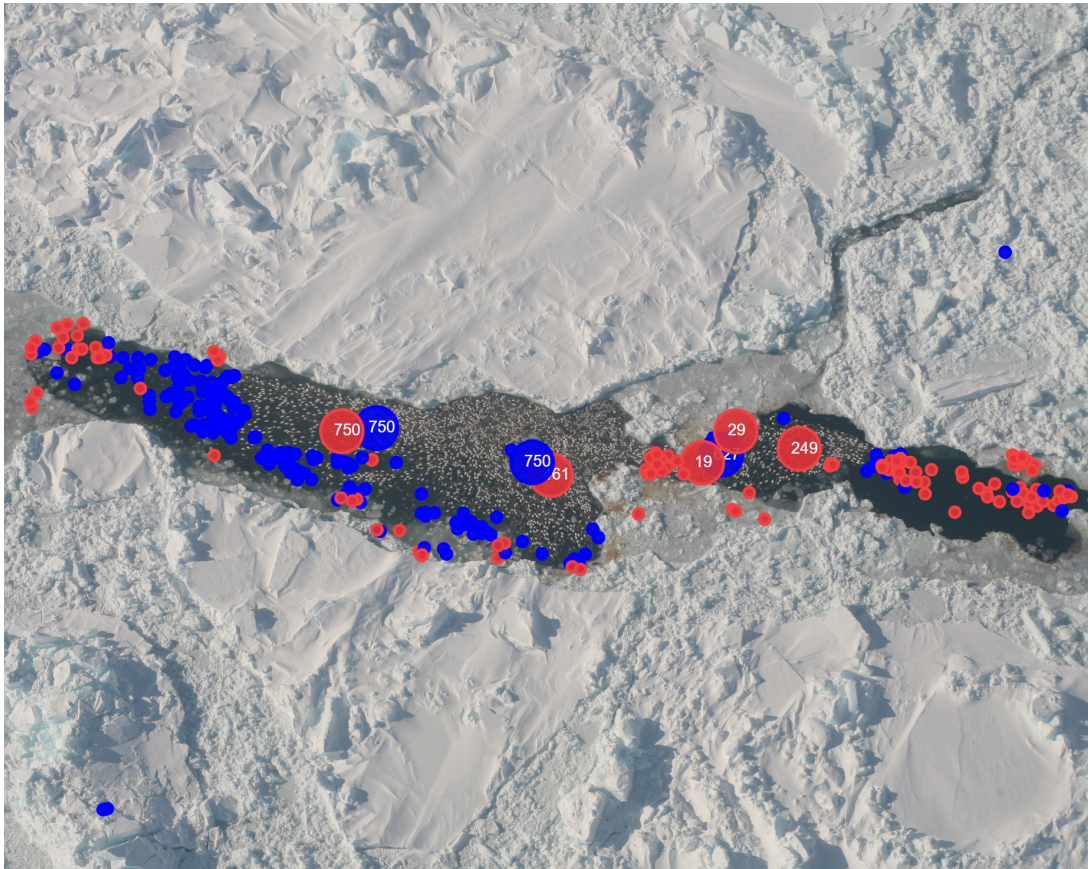
- Roboflow and CVAT do not support incremental re-training of a model.
- LabelStudio supports re-training, but it assumes that the image is fully-labeled.
 - For dense wildlife imagery, fully correcting model predictions before re-training is time-consuming.



Methodology



Methodology – Quality-of-life Features



Extra Model Parameters

Number of Epochs
Number of epochs for training the model.

4

Patch Size for training
Patch size for training the model.

Patch Size for predict
Patch size for inference from the model.

Batch Size for training
Batch size for training the model.

2

Batch Size for Slicing
For big images, we slice the images, and this parameter determines how many slices are sent to the model at once

4

- Dynamic bubbling based on proximity (Hierarchical-DBSCAN)
- Model checkpointing, weight uploads, and training parameter changes from UI

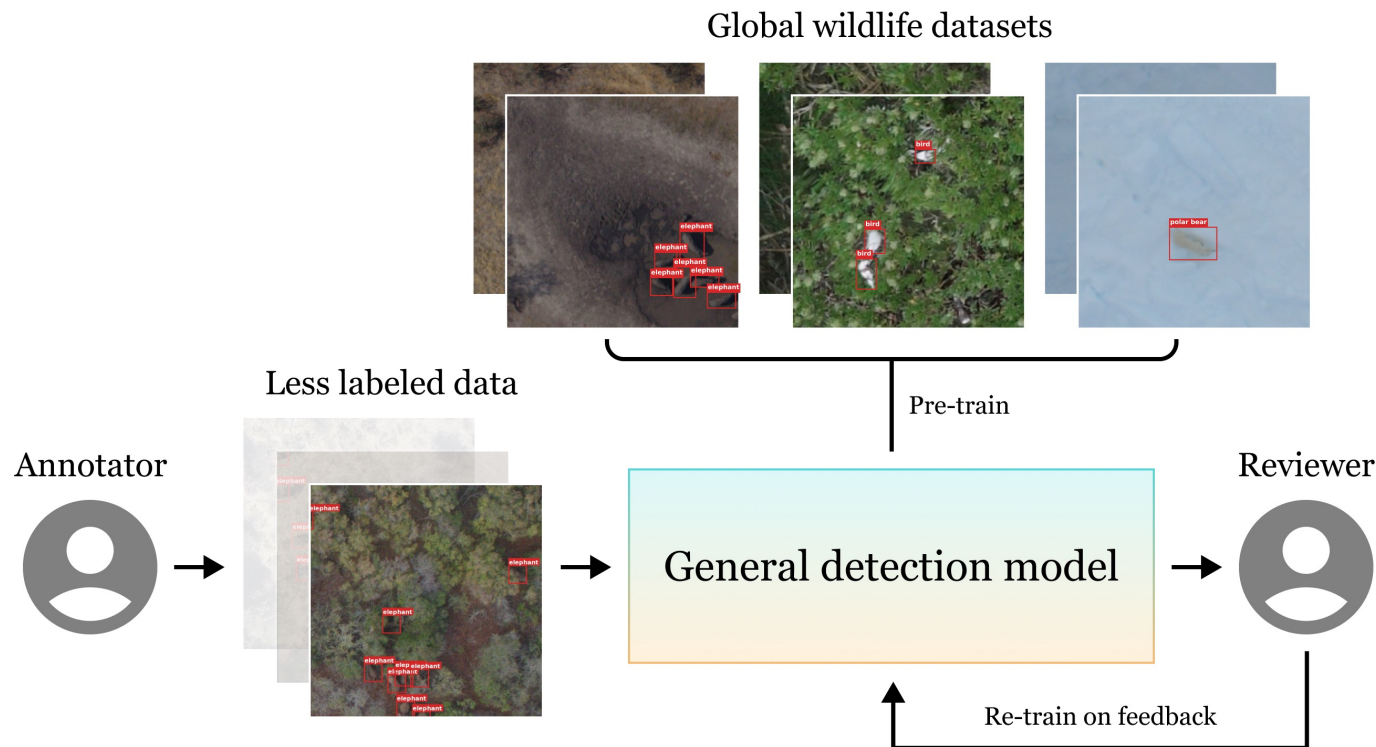
CONTRIBUTION 3

A case study with the Arctic Eider Society demonstrating the effectiveness of these components in real-world ecological monitoring.

Collaborators: Aryan Kalia, Zhonghao Zhang, Hudson Sun, Muhammed Patel, Becky Segal, Joel Heath

Research Question

- Does the combination of OpenWildlife and our custom LabelStudio interface improve real-world wildlife conservation efforts at the Arctic Eider Society?



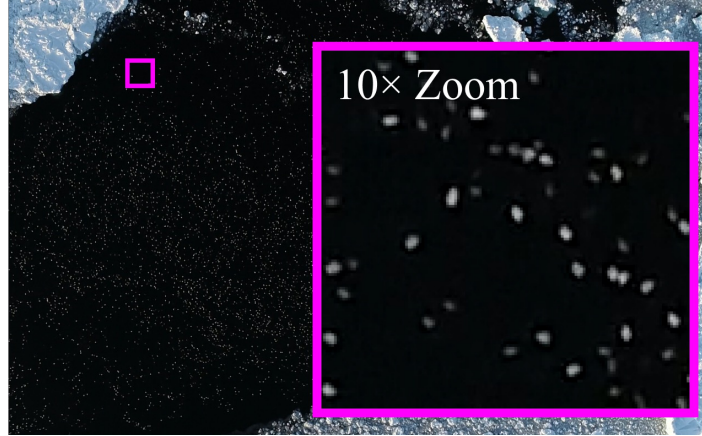
Age Group	Percentage
18-24	25%
25-34	30%
35-44	20%
45-54	15%
55-64	10%

-
- | Percentage | Color |
|------------|------------------|
| 45% | Light Purple |
| 35% | Medium Purple |
| 25% | Dark Purple |
| 15% | Very Dark Purple |



Dataset

- AES was founded to assess the impact of climate change on eiders.
- Backlog of 750 unanalyzed survey images from 2002, 2008, and 2021.
- Train: six images (8,339 annotations), took AES co-op *four months*.
- Test: 10 images (35,000 annotations), took us two weeks (with model).



Results – Accuracy of HITL Training

🤔 Does each successive step of HITL training improve model performance on our test dataset?

✅ Yes, the accuracy on the test dataset improves, up to a 22% error from the GT.

Pre-trained: model trained only on 15 aerial wildlife datasets following OpenWildlife pre-training.

Fine-tuned: pre-trained model further fine-tuned using the 6 fully annotated eider colony images.

HITL: fine-tuned model further refined by incorporating incremental human feedback using the annotation interface on 5 separate images.

Pre-trained	✓	✓	✓
Fine-tuned	✗	✓	✓
HITL	✗	✗	✓
Precision ↑	0.0	67.11	65.97
Recall ↑	0.0	75.64	77.56
MAE ↓	3384.7	548	264.3
% Error ↓	98.08	29.20	22.23

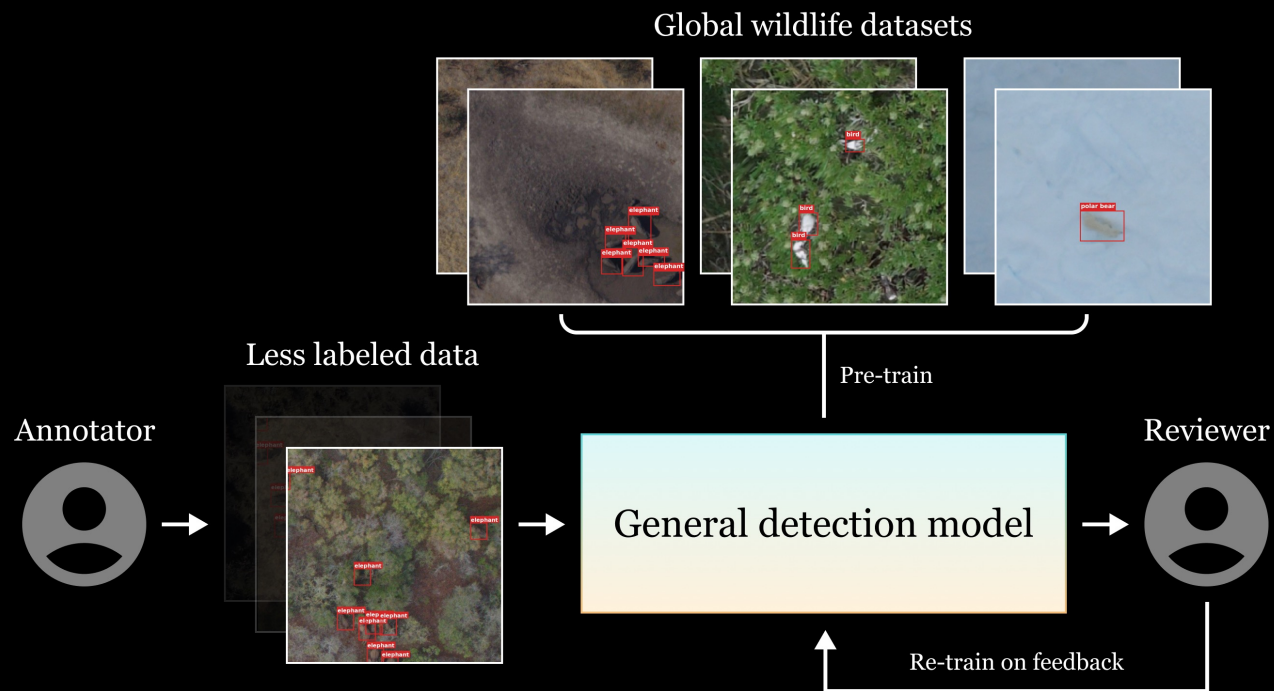
Results – Efficiency of HITL Workflow

🤔 Does the HITL workflow improve the efficiency of annotation?

✅ Yes, the HITL interface reduces the time required to annotate a single image by 87.5%.

Workflow	Time Per Image (minutes) ↓
Manual	117.08 ± 111.29
HITL	14.70 ± 9.73

CONCLUSION



Contributions:

1. A multi-species, language-enabled wildlife detection model supporting open-vocabulary identification.
2. A human-in-the-loop annotation workflow allowing incremental fine-tuning from user corrections.
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